

Ambiguity as Evidence: probabilistic preprocessing and topic uncertainty quantification in Ancient Greek philosophical texts

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Abstract

Ancient Greek is highly inflected, individual surface forms frequently admit several possible lemmas, and deterministic lemmatisation pipelines discard that ambiguity before any model sees the text. This paper treats morphological analysis as a probabilistic inference problem and propagates the resulting uncertainty through a topic model of twelve works of Plato and Aristotle. Three document-term matrices are constructed from the same corpus, encoding point estimation, posterior expectation, and full posterior sampling respectively, and Latent Dirichlet Allocation is fitted under each. Preprocessing uncertainty propagates into the fitted topic-word distributions, though a control separating morphological resampling from the fitter’s own initialisation variance shows the two to be of comparable magnitude: morphological propagation alone accounts for $\rho = 28.6\%$, against 36.88% for the confounded measure. What remains is structured rather than random, and sharper than the confounded measure indicated. The most morphologically stable topic sits 54% below the next once estimation noise is removed, and morphological instability predicts variance in topic coherence at $r = 0.90$ once coherence is computed on aligned topics. Instability varies systematically across topics, and within-work entropy varies with rhetorical register even where corpus size is fixed by construction. Extending the framework to four books of the Septuagint, a cosine-based vocabulary analysis separates genuine lexical overlap with the Platonic corpus from topic activation driven by thematic adjacency alone.

1 Introduction

Computational approaches to ancient texts have expanded rapidly, yet a 2019 survey found that nearly three quarters of LDA studies in digital humanities did not report their preprocessing procedures (Du, 2019). For Ancient Greek this is not a minor methodological detail. Greek morphology is highly inflected, individual surface forms often admit multiple possible lemmas, and deterministic lemmatisation pipelines typically discard this ambiguity entirely. Köntges (2020) argues that such ambiguity is primarily a corpus-size problem, one that becomes manageable through frequency as datasets grow larger. This paper argues instead that morphological ambiguity in Ancient Greek is semantically structured and varies systematically with rhetorical register, i.e. that explicit probabilistic treatment is methodologically necessary rather than merely fastidious.

To investigate this question we construct a Bayesian NLP pipeline for topic modelling twelve works of Plato and Aristotle (1,133 chunks, 1,336 lemmas), treating morphological analysis as a probabilistic inference problem and propagating uncertainty through the topic modelling pipeline. Rather than relying upon a single deterministic document-term matrix (DTM), we compare three representations: a baseline maximum a posteriori (MAP) DTM, a posterior-weighted DTM, and an ensemble of Monte Carlo sampled DTMs drawn from the full morphological posterior. Latent Dirichlet Allocation (Blei et al., 2003) is then fitted under each representation in order to measure how preprocessing uncertainty affects downstream topic structure. The primary task is cluster analysis: LDA identifies latent topical clusters across the corpus, and the analysis evaluates how

robustly those clusters are recovered under morphological uncertainty. A secondary regression task examines how topic prevalence varies with document-level covariates — author, period, and genre — using ordinary least squares upon the fitted topic proportions.

The results indicate that morphological ambiguity propagates non-trivially into topic modelling outputs, and that this uncertainty is structured rather than random. Ambiguity varies systematically across topics, genres, and rhetorical registers, with substantial within-work entropy variation that cannot be explained by corpus size. Extending the framework to the Septuagint further shows that topic activation in held-out texts can be separated into genuine lexical overlap and broader thematic similarity through cosine-based vocabulary analysis. Together these findings suggest that probabilistic preprocessing is an important condition for reliable downstream interpretation rather than a technical refinement.

2 Data and Preprocessing

2.1 Corpus

The training corpus consists of twelve works drawn from the Platonic and Aristotelian traditions: nine dialogues of Plato (*Republic*, *Symposium*, *Phaedrus*, *Phaedo*, *Sophist*, *Ion*, *Meno*, *Apology*, *Timaeus*) and three treatises of Aristotle (*Metaphysics*, *Nicomachean Ethics*, *Poetics*), spanning early, middle, and late Platonic periods and covering the range of Aristotelian philosophical genres. Texts were scraped from the Perseus Digital Library via the Scaife CTS API and segmented into 300-token chunks, yielding 1,133 training documents. After vocabulary filtering ($\text{min_df} = 5$, $\text{max_df} = 0.9$), the training vocabulary contains 1,336 lemmas.

Four books of the Septuagint (Genesis, Proverbs, Wisdom of Solomon, Sirach) serve as the held-out projection corpus, yielding 227 additional chunks. The Septuagint is a useful held-out test because it crosses a cultural, linguistic, and religious boundary: composed under direct Hellenistic philosophical influence, it is a natural test case for whether Platonic vocabulary penetrated Greek Jewish scripture.

Morphological analysis was performed via the Perseids Morpheus API, a rule-based parser for Ancient Greek that returns a posterior distribution over possible lemmas for each surface form. The full corpus required 44,879 cached morphological analyses. Morpheus returns multiple candidate lemmas with associated probabilities for ambiguous forms, while unambiguous forms receive a point mass upon a single lemma. This distributional output is the raw material for the three DTM constructions described below.

2.2 Morphological entropy as a standalone result

Before constructing any topic model, the morphological posterior distributions are themselves informative. For each surface form w , let $p_1, \dots, p_m$ denote the posterior probabilities over its m candidate lemmas. The lemma entropy is $H(w) = -\sum_{i=1}^m p_i \log_2 p_i$, with $H(w) = 0$ for unambiguous forms. Mean lemma entropy per work, averaging $H(w)$ over all unique surface forms, is reported in Table 1.

Two patterns are visible in Table 1, and one complication that bears upon both.

The first pattern is authorial. The three Aristotelian works occupy the three lowest entropy positions in the corpus, while the Platonic works range from *Ion*, at 0.244 bits, down to *Republic*, at 0.194; if author were unrelated to entropy, the probability that three particular works would occupy the three lowest ranks of twelve is $1/\binom{12}{3}$, i.e. 0.005. The second is that entropy varies within Plato and not only between the two authors, since *Ion*, at the upper bound of the Platonic range, is 25.8% more ambiguous than *Republic* at the lower.

However, both patterns are confounded with corpus size, and it is better to concede this at the outset than to leave it to be discovered. Across the twelve works, mean entropy falls with log token count at $r = -0.72$ ($p = 0.009$), and the relationship is no weaker within the Platonic corpus considered alone ($r = -0.72$,

Table 1: Mean lemma entropy by work, ordered by entropy.

Work	Tokens	Mean H (bits)	Ambiguous forms
Ion	3,652	0.244	260/1,310
Meno	9,310	0.233	443/2,362
Phaedrus	16,066	0.232	999/5,120
Phaedo	21,196	0.209	863/4,874
Timaeus	23,316	0.208	1,147/6,403
Symposium	16,777	0.206	824/4,683
Apology	8,449	0.205	443/2,599
Sophist	15,712	0.200	638/3,824
Republic	85,415	0.194	2,550/15,063
Poetics	9,691	0.192	452/2,797
Nicomachean Ethics	54,202	0.186	1,370/8,506
Metaphysics	75,613	0.185	1,096/6,876

$p = 0.027$); regressing entropy upon log tokens together with an author indicator, the coefficient upon size remains significant ($p = 0.026$) while the author term does not ($p = 0.072$).¹ Smaller works are, on average, the more morphologically ambiguous, which is precisely what Köntges (2020) predicts.

Two size-matched comparisons suggest that the author effect is nonetheless present. *Meno* (9,310 tokens, $H = 0.233$) and *Poetics* (9,691 tokens, $H = 0.192$) differ in length by roughly four percent, yet the Platonic work is 21% the more ambiguous; the same ordering holds at the other extreme of the range, where *Metaphysics* (75,613 tokens, $H = 0.185$) sits below *Republic* (85,415 tokens, $H = 0.194$). With twelve works, however, the two effects cannot be separated convincingly, and we do not claim to have separated them. The work-level table is not where this question is settled. Settling it requires that corpus size be held exactly fixed, which is what the within-work analysis of Section 5.1 does.

2.3 Three-method DTM construction

The central preprocessing contribution of this paper is the construction of three document-term matrices from the same underlying corpus, each encoding a different assumption about how morphological uncertainty should be handled.

The **baseline** DTM assigns each token its maximum a posteriori (MAP) lemma, i.e. the single most probable candidate under the Morpheus posterior. This is the standard approach in the digital humanities literature, including Köntges (2020), and it discards all distributional information about morphological ambiguity.

The **weighted** DTM replaces each token’s MAP lemma with a fractional count distributed across all candidate lemmas proportionally to their posterior probabilities. A token with posterior ($p_1 = 0.7, p_2 = 0.3$) over two lemmas contributes 0.7 to the first lemma’s count and 0.3 to the second’s, rather than 1.0 to the MAP lemma alone. This preserves the expected lemma distribution without introducing stochasticity.

The **sampled** DTM generates 50 independent realisations of the full corpus by drawing one lemma per token from its posterior distribution. Each realisation produces a distinct DTM, and LDA is fitted independently upon each. The ensemble of 50 fitted topic-word matrices $\{\phi^{(s)}\}_{s=1}^{50}$ then supports direct uncertainty quantification: for each topic-word pair (k, v) , the empirical standard deviation $\phi_{\text{std}}[k, v] = \text{std}(\{\phi^{(s)}[k, v]\})$

¹The regression is reported for completeness rather than as an estimate we lean upon. With twelve observations and two correlated regressors, neither coefficient is precisely determined, and the author term’s failure to reach conventional significance is as much a statement about the sample as about the effect.

measures how much morphological ambiguity propagates into the estimated topic-word distribution. This is the quantity that drives the primary results of Section 4.2.

The three constructions represent distinct epistemic stances toward morphological ambiguity — point estimation, posterior expectation, and full posterior sampling — and differences across their outputs quantify how much the downstream analysis depends upon uncertainty that is routinely ignored. Schofield and Mimno (2016) showed that preprocessing choices are inconsequential for LDA coherence in English, whereas May et al. (2016) showed that the opposite holds for morphologically rich languages. Our construction extends this by treating lemmatisation uncertainty itself as a measurable quantity: Vatri and McGillivray (2020) found that no current Greek lemmatiser provides reliable token-level confidence scores, a gap this pipeline addresses directly in the spirit of Denny and Spirling’s (2018) preprocessing sensitivity analysis.

3 Methods

3.1 Latent Dirichlet Allocation

Latent Dirichlet Allocation (LDA; Blei et al., 2003) represents each document as a mixture of K latent topics and each topic as a distribution over vocabulary items. Each document d has topic proportions $\theta_d \sim \text{Dirichlet}(\alpha)$ and each topic k has a word distribution $\phi_k \sim \text{Dirichlet}(\beta)$; tokens are generated by drawing a topic $z \sim \text{Categorical}(\theta_d)$ and then a word $w \sim \text{Categorical}(\phi_z)$. Here θ supports the author separation and periodization analysis, and ϕ supports the uncertainty quantification. We fit LDA using scikit-learn’s online variational Bayes implementation with $K = 15$ topics, $N_{\text{iter}} = 100$ iterations, and random seed 42. A grid search over $K \in \{10, 12, 15, 20\}$ confirmed that $K = 15$ maximises the number of philologically interpretable topics while avoiding fragmentation at higher values.²

3.2 Uncertainty propagation via ϕ_{std}

The central methodological contribution is the propagation of morphological uncertainty through the LDA fitting procedure. Under the sampled DTM construction of Section 2.3 we obtain 50 independent LDA fits $\{\phi^{(s)}\}_{s=1}^{50}$, one per sampled corpus realisation, and the primary uncertainty measure is

$$\phi_{\text{std}}[k, v] = \text{std}\left(\left\{\phi^{(s)}[k, v]\right\}_{s=1}^{50}\right).$$

This quantity measures, for each topic-word pair (k, v) , how much the estimated word probability in topic k varies across the 50 morphological realisations of the corpus. The ratio of the mean ϕ_{std} across all topic-word pairs to the standard deviation of the baseline ϕ provides a corpus-level summary of uncertainty propagation:

$$\rho = \frac{\mathbb{E}_{k,v}[\phi_{\text{std}}[k, v]]}{\text{std}(\phi_{\text{baseline}})}.$$

Under this procedure $\rho = 36.88\%$. That figure, however, does not isolate what it appears to isolate, and the qualification is substantial enough to state before any result is read.

Each of the fifty sampled fits is initialised with its own random state, so ϕ_{std} pools morphological resampling with the sensitivity of online variational Bayes to initialisation, i.e. with the tendency of the fitter to settle in different local optima from different starting points. We separate the two with a pair of matched controls at

²Priors are left at the scikit-learn defaults, $\alpha = \beta = 1/K$. Wallach et al. (2009) show that asymmetric priors over θ materially affect fitted topic structure, so this is a design choice rather than a neutral default, and one whose consequences we do not explore here.

the same $S = 50$, in which the reference topic ordering, the alignment step and the hyperparameters are held identical and only the source of variation differs:

Arm	Varies	ρ
Full	corpus realisation <i>and</i> random state	36.88%
Seed-only	random state alone, corpus fixed	36.44%
Morphology-only	corpus realisation alone, random state fixed	28.64%

The full arm is the published procedure: it recovers $\rho = 36.88\%$ and reproduces the per-topic values of Table 3 to within 5×10^{-6} , so the controls are directly comparable rather than approximately so. The consequential finding is that seed variation alone recovers 36.44%, i.e. 98.8% of the full spread. Estimation noise, rather than morphology, accounts for almost the whole of what ρ measures under the published protocol.

Morphological propagation is nonetheless real and substantial. Holding the random state fixed at 42 and varying only the corpus realisation gives $\rho_{\text{morph}} = 28.64\%$, and it is this figure rather than 36.88% that we report as the propagation of morphological uncertainty into the topic-word distributions. The two sources are sub-additive rather than independent, their quadrature sum being 46.35% against 36.88% observed, which is what one would expect where both perturb the same fitting problem.

3.3 Topic coherence

To assess whether morphologically unstable topics are also less coherent, we compute Normalised Pointwise Mutual Information (NPMI) coherence for each topic across the fifty fits of each arm described in Section 3.2. Coherence is computed on the *aligned* ϕ matrices, i.e. after the Hungarian matching that reconciles topic permutation across fits.³ For a topic with top- n words $w_1, \dots, w_n$,

$$C_{\text{NPMI}} = \frac{2}{n(n-1)} \sum_{i < j} \frac{\log \frac{p(w_i, w_j)}{p(w_i)p(w_j)}}{-\log p(w_i, w_j)},$$

where probabilities are estimated from document co-occurrence frequencies in the training DTM. The standard deviation of NPMI coherence across the fifty aligned fits of an arm, C_{std} , measures how much a topic’s semantic coherence varies under that arm’s source of perturbation. We test whether ϕ_{std} predicts C_{std} via Pearson correlation, within each arm, so that both quantities refer to the same topics and the same source of variation.

3.4 Projection of held-out works

To project the Septuagint into the trained topic space we refit LDA upon the baseline training DTM using scikit-learn’s online variational Bayes implementation, which exposes a `transform` method for out-of-sample documents. Each held-out chunk is represented as a vocabulary-aligned DTM row — counts over the 1,336 training lemmas, with out-of-vocabulary terms discarded — and transformed into a topic proportion vector θ_{new} using the fitted model’s topic-word distributions. The mean θ across all chunks of a given work gives the work-level topic profile used in the projection heatmap.

To distinguish genuine philosophical vocabulary overlap from topic-activation false positives we supplement the θ -level projection with a ϕ -level cosine analysis. For each Septuagint work and focus topic we identify the

³This corrects an error in the original analysis, where coherence was computed on unaligned fits. Because LDA does not preserve topic ordering across runs, the resulting $C_{\text{std}}[k]$ pooled the coherences of whatever topic occupied slot k in each fit and was therefore not a per-topic quantity at all. Section 4.2 reports what the correction changes.

top-scoring chunks by topic weight, extract their raw term frequency vectors restricted to the top 20 topic words, and compute the cosine similarity between the Septuagint mean vector and the corresponding pooled Platonic reference vector. A cosine above 0.6 is classified as genuine shared vocabulary; between 0.3 and 0.6 as ambiguous; below 0.3 as a false positive driven by topic overlap rather than lexical correspondence.

3.5 Register entropy and covariate regression

For the register entropy analysis we compute per-chunk token-weighted entropy, i.e. the mean $H(w)$ averaging over all token positions rather than over unique surface forms, weighting ambiguous high-frequency forms proportionally to their contribution to the DTM.

For the covariate regression we regress chunk-level topic proportions $\theta_{d,k}$ upon document metadata using ordinary least squares:

$$\theta_{d,k} = \beta_0 + \beta_1 \cdot \text{author}_d + \beta_2 \cdot \text{period}_d + \beta_3 \cdot \text{genre}_d + \varepsilon_d,$$

where author is a binary indicator (Plato/Aristotle), period is categorical (early, middle, late Plato, Aristotle), and genre is categorical (dialogue, treatise, drama criticism). This is a post-hoc structural topic model in the spirit of Roberts et al. (2014), without the joint estimation that a full STM would provide. The R^2 of each per-topic regression measures how much of the variance in topic prevalence is explained by author, period, and genre metadata.

4 Results

4.1 Topic structure

Table 2 presents the ten topics most central to the results discussion, labelled by inspection of the top words under ϕ_{baseline} . Labels are interpretable against the philological literature, and the overall topic organisation is robust to morphological uncertainty, with the quantified divergence captured by ϕ_{std} in Section 4.2.⁴

4.2 Morphological uncertainty propagation

The headline quantitative result must be read through the decomposition of Section 3.2: morphological propagation accounts for $\rho_{\text{morph}} = 28.64\%$, against 36.88% for the confounded measure and 36.44% for estimation noise alone. The uncertainty that remains is not uniform across topics, and the decomposition changes materially which topics carry it. Table 3 reports mean ϕ_{std} under the published procedure alongside all three controls.

Two things follow. The first is that T7 (Metaphysics/Ontology) is the most stable topic under every arm, and that its stability is specifically morphological. Under the morphology-only control it sits 54% below the next-lowest topic, against 11% under the full procedure; its morphological share of 0.40 is the lowest in the corpus. The confounded measure was concealing this result rather than producing it. The contrast with T12 (Form/Beauty) sharpens correspondingly, from a ratio of 3.19 under the published procedure to 5.82 once estimation noise is removed. T7 is anchored by the dense technical vocabulary characteristic of Aristotelian

⁴The top-word lists are filtered for display. The raw ϕ ranking includes elided particles and homograph-numbered lemmas — οὐδ', ἄνω1, ἄπειρος1 — which survived stopword removal. Their presence is itself a small piece of evidence for the argument of this paper, since deterministic preprocessing leaves residue even where it is carefully specified; Schofield, Magnusson and Mimno (2017) show that stopword removal is consequential for fitted topic structure and poorly served by standard lists, which is the condition we observe here.

Table 2: Ten topics central to the results discussion. Topics T1, T2, T4, T6 and T13 are omitted for brevity; the full structure is available in the supplementary notebook.

Topic	Label	Top words
T0	Mimesis/Poetics	τραγωδία, μῦθος, μίμησις, λέξις, κωμωδία
T3	Division/Classification	γένος, εἶδος, διαφορά, ζῴιον, διαιρέω
T5	Pleasure/Friendship	ἡδονή, φίλος, λύπη, χαίρω, εὐδαίμων
T7	Metaphysics/Ontology	οὐσία, ἀριθμός, κίνησις, αἰσθητός, λευκός
T8	Law/Character	νόμος, φύλαξ, ἥθος, ἀμαρτάνω, τιμάω
T9	Justice/Virtue	δικαιοσύνη, ἀδικία, ἔξις, ἀρετή
T10	Soul/Death	σῶμα, ἀποθνήσκω, ἀθάνατος, θάνατος
T11	Teaching/Inquiry	ἡγέομαι, πείθω, γράφω, νομίζω, διδάσκω
T12	Form/Beauty	κάλλος, σφαῖρα, ἀνάλογος, χρῶμα
T14	Elements/Cosmogony	σῶμα, πῦρ, Γαῖα, ὕδωρ, γένεσις, οὐρανός

exposition — οὐσία, ἀριθμός, κίνησις — terms that recur with relatively stable surface forms across the treatises, and it is precisely this that the clean measure detects.

Three topics carry a morphological share above unity: T1 and T8 at 1.05, and T11 at 1.07. This is not an arithmetic error but a consequence of the sub-additivity noted in Section 3.2. Because the two perturbations act upon the same fitting problem rather than upon independent quantities, their combined effect is smaller than the quadrature sum, and for a topic where they interact destructively the full arm can fall marginally below the morphology-only arm. The three cases sit within seven per cent of unity and we read them as noise about a share of one, i.e. as topics whose instability is morphological in essentially its entirety.

The second is that T12 is no longer the most unstable topic. Under the morphology-only control that position is held by T1 (Cosmology/Vision) at 0.00196, with T0 second and T12 third at 0.00160. T12 remains within the top three, and the reading it invites is unaffected in kind: its vocabulary — κάλλος (beauty), σφαῖρα (sphere), ἀνάλογος (proportional), χρῶμα (colour) — spans aesthetic, geometric and perceptual registers simultaneously, which makes it inherently sensitive to how ambiguity is resolved. The superlative, however, does not survive, and several topics that appeared unstable under the published measure prove substantially to have been artefacts of initialisation: T13 (Music/Mathematics) falls from fifth to tenth, and T7, T3, T10 and T13 each derive little more than half of their apparent instability from morphology. Across the fifteen topics the full and morphology-only orderings correlate at $r = 0.797$ (rank 0.786), so the ordering is attenuated rather than destroyed; but the published ordering tracks seed sensitivity far more closely still ($r = 0.965$, rank 0.954), which is the clearest single statement of what the confound was doing.

The ordering by mean conceals a further wrinkle worth naming. T7 carries the lowest mean ϕ_{std} in the corpus yet, under the published procedure, a maximum of 0.046, higher than the 0.037 of T12; a topic may be stable across nearly the whole of its vocabulary while one or two of its words move a great deal. Mean ϕ_{std} measures the former. It is the right summary for asking whether a topic’s overall shape is robust, and the wrong one for asking whether any particular word within it is.

The relationship between morphological instability and coherence variance is considerably stronger than the original analysis reported, and it strengthens for two reasons that compound. Computing coherence on aligned topics within the morphology-only arm, mean ϕ_{std} per topic correlates with C_{std} at $r = 0.902$ (rank 0.914, $p < 10^{-4}$), against $r = 0.545$ ($p = 0.036$) as originally reported. The corrected correlation is robust to the removal of any single topic, the leave-one-out minimum being $r = 0.872$, where the original figure fell to $r = 0.349$ if T12 alone was dropped.

Table 3: Mean ϕ_{std} by topic, ordered by the morphology-only control. The published column is the original run; the full arm reproduces it to within 5×10^{-6} . All arms use $S = 50$. Morphological share is the ratio of the morphology-only arm to the full arm.

Topic	Label	Published	Full	Seed-only	Morph-only	Morph. share
T1	Cosmology/Vision	0.00187	0.00187	0.00192	0.00196	1.05
T0	Mimesis/Poetics	0.00176	0.00176	0.00152	0.00163	0.93
T12	Form/Beauty	0.00221	0.00221	0.00210	0.00160	0.73
T4	Dialectic/Falsehood	0.00181	0.00181	0.00186	0.00151	0.84
T9	Justice/Virtue	0.00150	0.00150	0.00153	0.00127	0.85
T6	Desire/Eros	0.00142	0.00142	0.00136	0.00126	0.89
T2	Forms/Knowledge	0.00161	0.00161	0.00175	0.00111	0.69
T8	Law/Character	0.00098	0.00098	0.00115	0.00103	1.05
T5	Pleasure/Friendship	0.00095	0.00095	0.00077	0.00092	0.97
T13	Music/Mathematics	0.00171	0.00171	0.00174	0.00091	0.53
T11	Teaching/Inquiry	0.00080	0.00080	0.00088	0.00086	1.07
T14	Elements/Cosmogony	0.00109	0.00109	0.00102	0.00065	0.60
T10	Soul/Death	0.00124	0.00124	0.00126	0.00064	0.52
T3	Division/Classification	0.00127	0.00127	0.00110	0.00060	0.47
T7	Metaphysics/Ontology	0.00069	0.00069	0.00068	0.00028	0.40

It is also ordered across the three arms in the way a morphological reading predicts: strongest where only the corpus varies ($r = 0.902$), weaker under the full procedure ($r = 0.684$), and weakest where only the random state varies ($r = 0.557$). Had the association been an artefact of estimation noise, the ordering would run the other way. Topics whose word distributions are sensitive to how morphological ambiguity is resolved are also topics whose semantic coherence moves under resampling, and the effect is large enough that fifteen topics suffice to detect it.

The original figure understated the relationship because of an error in the coherence pipeline rather than a property of the data. Coherence was computed on unaligned fits, so $C_{\text{std}}[k]$ pooled the coherences of whichever topic occupied slot k in each fit, attenuating the association toward zero. Correcting the alignment and isolating morphology together take the result from marginal and single-topic-dependent to robust. That the confound and the alignment error worked in opposite directions — one inflating ρ , the other suppressing the coherence correlation — is a reminder that the direction of a methodological error is not predictable from its presence.

What does not change is the negative result. There is no relationship between morphological instability and *mean* coherence under any arm ($r = -0.285$ under morphology-only, $p = 0.30$; $r = -0.059$ under the full procedure). Morphologically unstable topics are not less coherent on average. They are topics whose coherence is less stable, which is a different and more interesting claim.

4.3 Author separation and periodization

The mean document-topic weight θ across the twelve training works separates Plato from Aristotle at a glance. The three Aristotelian works show markedly higher weight upon the metaphysics and classification topics, T7 and T3, while the Platonic works concentrate weight upon T10, T11 and T8, i.e. soul and death, teaching and inquiry, law and character. This author-level separation is confirmed by hierarchical clustering, which uses Jensen–Shannon divergence between work-level θ distributions as a distance metric under Ward linkage. Plato and Aristotle form cleanly separated clusters with no cross-author merging at any level of the dendrogram.

Within Plato, the dendrogram recovers a structure broadly consistent with, though not identical to, the stylometric periodization established by Brandwood (1992): the early dialogues (*Apology*, *Meno*, *Ion*) form a dense sub-cluster, the middle dialogues (*Republic*, *Symposium*, *Phaedrus*, *Phaedo*) form a second, and the late dialogues (*Sophist*, *Timaeus*) cluster together before merging with the rest of Plato. A t-SNE projection confirms this at the chunk level, with Platonic chunks showing a gradient from early to late along the primary axis of variation that the dendrogram captures only coarsely. Storey and Mimno (2020), mapping stylistic similarity across the full Ancient Greek corpus via Jensen–Shannon divergence, similarly find that Plato occupies a distinct and internally structured region of the stylistic space, a result our topic-based analysis recovers independently.

4.4 Republic book-level analysis

Disaggregating the *Republic* by book group reveals within-work topical variation that the work-level analysis conceals. T3 (Division/Classification) was excluded from this analysis because its vocabulary — γένος (genus), εἶδος (species), διαφορά (difference) — appears as background argumentative scaffolding uniformly across all book groups, suppressing variation in the topics of substantive interest; topic proportions were renormalised after exclusion.

Across all six book groups, T10 (Soul/Death) and T11 (Teaching/Inquiry) remain dominant, reflecting the *Republic*’s sustained engagement with eschatology and Socratic pedagogy. The highest T10 weight appears in Books 8–9 (Decline, $\bar{\theta} = 0.304$), consistent with the sustained treatment of the soul’s degradation under tyranny. Book 10 is the only section to show a signal from T5 (Pleasure/Friendship, $\bar{\theta} = 0.152$), corresponding to the closing myth of Er — a topical shift the model detects without supervision, indicating that the topic structure recovers genuine textual content rather than artefacts of corpus construction.

5 Extensions

5.1 Register entropy as a rebuttal to Köntges

Köntges (2020) argues that morphological complexity in Ancient Greek is a corpus-size problem, on the grounds that larger corpora allow frequency-based disambiguation to overcome morphological ambiguity. The within-work entropy analysis offers a sharp corrective: if ambiguity were strictly a function of corpus size, entropy should be uniform within a single work, but instead it ebbs and flows with the text’s rhetorical purpose. Table 4 reports section-level mean token-weighted entropy for four works chosen to span the range of compositional structure in the corpus.

Since corpus size is fixed within a single work, any entropy variation across sections must reflect something other than data quantity. The *Phaedo* is the most compelling case: the afterlife myth (mean $H = 0.2062$) is conspicuously less ambiguous than the dialectical arguments (mean $H = 0.2495$), a difference of 19.6% of the work’s section mean. Mythological narrative draws upon a simpler and more constrained morphological register than dialectical argument, with shorter sentences, less subordination, and fewer rare verb forms; the afterlife myths of Plato are compositionally distinct passages deploying a different vocabulary, and the entropy signal distinguishes them from the surrounding dialectical text without any supervision. The *Metaphysics* shows a complementary pattern: Books 5–6, which function as a philosophical lexicon defining technical terms, show the highest entropy (0.2525), while the more systematic Books 10–11 show the lowest (0.2129). Definitional prose, which introduces rare technical vocabulary, is morphologically more ambiguous than systematic argument that reuses established terms.

The *Timaeus* result runs in the opposite direction and is itself a positive finding. Its within-work range of 4.7% is the flattest profile in the corpus. Unlike the *Phaedo* or the *Republic*, the *Timaeus* is a sustained technical

Table 4: Section-level token-weighted entropy within four works. Ranges are expressed relative to the mean of that work’s section means, both quantities being token-weighted.

Work	Section	Mean H	n chunks
Phaedo	Biographical frame (57–64)	0.2525	8
	Arguments (64–107)	0.2495	54
	Afterlife myth (107–118)	0.2062	9
	<i>within-work range: 0.0463, i.e. 19.6% of the section mean</i>		
Metaphysics	Books 1–2: Historical	0.2209	36
	Books 3–4: Aporetic	0.2228	36
	Books 5–6: Lexicon/Science	0.2525	36
	Books 7–9: Substance	0.2410	72
	Books 10–11: Unity/Science	0.2129	36
	Books 12–14: Theology	0.2279	36
	<i>within-work range: 0.0396, i.e. 17.2% of the section mean</i>		
Republic	Book 1: Justice	0.2539	26
	Books 2–3: Education	0.2571	61
	Book 4: Virtues	0.2650	24
	Books 5–7: Philosopher	0.2565	87
	Books 8–9: Decline	0.2352	47
	Book 10: Mimesis/Myth	0.2522	40
	<i>within-work range: 0.0298, i.e. 11.8% of the section mean</i>		
Timaeus	Prologue (17–27)	0.2145	10
	Cosmology (27–47)	0.2198	30
	Second cosmology (47–69)	0.2207	20
	Physiology (69–92)	0.2106	18
	<i>within-work range: 0.0101, i.e. 4.7% of the section mean</i>		

monologue without the register shifts that characterise dialogic or multi-genre works, and the uniform entropy profile is what the register hypothesis predicts for a work of this compositional character.⁵

Together these results indicate that morphological ambiguity in Ancient Greek is not simply a shortage of data to be remedied by more text; it varies systematically with what a passage is doing, and the appropriate methodological response is to model that variation rather than resolve it away, which is what the three-DTM construction of Section 2.3 does.

5.2 Septuagint projection

All four Septuagint works show substantial weight upon T10 (Soul/Death), and three of the four upon T8 (Law/Character); in Wisdom, T11 and T14 displace it. This reflects the thanatological and legal-ethical vocabulary shared between Greek philosophical prose and the Septuagint’s wisdom and narrative literature. The more discriminating question is whether this topic activation is driven by genuinely Platonic vocabulary or by general Greek vocabulary that happens to co-occur with philosophical themes. Table 5 addresses this directly.

The most consequential result is Wisdom T14 = 0.8105. The Wisdom of Solomon — the most Hellenised

⁵We note that this reading was arrived at after the profiles had been inspected, and we therefore treat the *Timaeus* result as an observation consistent with the register hypothesis rather than as a test of it. A pre-registered assignment of works to register categories would convert it into one.

Table 5: Selected cosine similarities between Septuagint and pooled Platonic top-chunk vocabulary vectors. Full results are available in the supplementary notebook.

Work	Topic		Cosine	Classification
Wisdom	T14	Elements/Cosmogony	0.8105	Genuine
Sirach	T14	Elements/Cosmogony	0.6993	Genuine
Proverbs	T9	Justice/Virtue	0.6052	Genuine
Genesis	T14	Elements/Cosmogony	0.4791	Ambiguous
Genesis	T10	Soul/Death	0.4625	Ambiguous
Wisdom	T10	Soul/Death	0.3324	Ambiguous
Genesis	T13	Music/Mathematics	0.2947	False positive
Sirach	T10	Soul/Death	0.2924	False positive

book of the Septuagint, composed in Greek rather than translated from Hebrew — shows genuine lexical overlap with Platonic cosmological vocabulary. This is consistent with the scholarly consensus that the Wisdom of Solomon represents the deepest point of Platonic influence in the Septuagint corpus, and provides a quantitative anchor for that claim. Sirach T14 = 0.6993 crosses the genuine threshold by a smaller margin, consistent with Sirach’s more limited direct engagement with Hellenistic philosophy.

At first glance the Genesis result looks comparable, since Genesis scores highest upon T14 among the Septuagint works (mean $\theta = 0.185$) and the creation narrative shares obvious thematic ground with Platonic cosmogony. The cosine analysis complicates this picture: $\text{cosine}(\textit{Timaeus}, \text{Genesis}) = 0.496$ against $\text{cosine}(\text{other Plato}, \text{Genesis}) = 0.625$, so that Genesis is more similar to non-cosmological Platonic vocabulary than to the *Timaeus* specifically. The shared vocabulary driving the T14 activation — Γαῖα (earth), ὕδωρ (water), γένος (kind), οὐρανός (heaven) — appears across the Platonic corpus generally rather than in terms specific to the *Timaeus*, and the activation is therefore a thematic echo rather than a literary debt. The two false positive cases confirm the method’s discriminating power: Genesis T13 is driven by genealogical numerical vocabulary and Sirach T10 by general thanatological discourse, both thematically adjacent to their Platonic counterparts but lexically distinct.

Wisdom’s activation of T10 requires a more careful reading than the others, and the distinction upon which it turns is worth stating plainly. Its cosine against the pooled Platonic reference is 0.3324, which falls within the ambiguous band, and the vocabulary comparison explains why: Wisdom shares the general terms for death and for living — σῶμα, ζάω, βίος, θάνατος — while the vocabulary specific to the Platonic treatment of the soul’s immortality is absent, since ἀποθνήσκω occurs 1.4 times per top chunk in the *Phaedo* and not once in Wisdom’s top chunks. Narrowing the comparison to the *Phaedo* alone raises the cosine to 0.6775.⁶ However, that figure is not on the same scale as those reported in Table 5 and should not be read against its thresholds; we report it only to indicate that Wisdom’s thanatological register sits nearer to the *Phaedo* than to Platonic usage at large. What Wisdom’s activation of T10 reflects, then, is a shared vocabulary for death rather than the absorption of a Platonic doctrine of the soul.

⁶Two distinct quantities are in play, and conflating them would be an error. The cosines reported in Table 5, and the thresholds of 0.3 and 0.6 applied to them, compare each Septuagint work’s top-chunk vocabulary vector against a *pooled* Platonic reference constructed from the training corpus. The figure of 0.6775 compares Wisdom against the *Phaedo* alone. A work-specific comparison will generally exceed the pooled one wherever the held-out text resembles a particular source more closely than the tradition in aggregate, i.e. precisely the situation the pooled threshold is designed to detect and the work-specific figure is not calibrated to adjudicate.

Table 6: Within-Plato periodization regression: top five topics by R^2 . The reference category is early Plato. Full results are available in the supplementary notebook.

Topic	Label	R^2	Coef: middle	Coef: late
T14	Elements/Cosmogony	0.322	0.031	0.216
T11	Teaching/Inquiry	0.121	-0.236	-0.254
T3	Division/Classification	0.067	0.009	0.037
T10	Soul/Death	0.064	0.070	-0.067
T8	Law/Character	0.056	0.056	-0.018

5.3 Covariate regression

Table 6 reports the five topics with highest R^2 from the within-Plato periodization regression, which isolates the effect of period upon topic prevalence within the Platonic corpus alone.

T14 (Elements/Cosmogony) shows the strongest periodization signal ($R^2 = 0.322$, coefficient upon late = 0.216), driven by the concentration of cosmological vocabulary in the late Platonic corpus. T11 (Teaching/Inquiry) shows the second strongest signal and the more interpretable coefficient: the negative late coefficient (-0.254) reflects the structural shift from Socratic dialogue, built around question and answer, refutation and inquiry, to the monological expository prose of the late works. Early Plato is structurally constituted by the mechanics of Socratic questioning; late Plato replaces this with sustained argument. The topic model detects this transition without access to any metadata about dialogue structure, the metadata entering only to label the periods after the fact.

The full metadata regression including Aristotle reveals that genre is the strongest predictor of topic prevalence overall: T7 (Metaphysics/Ontology, $R^2 = 0.364$) and T0 (Mimesis/Poetics, $R^2 = 0.333$) are more strongly predicted by genre than any topic is by period, though the margin over T11 ($R^2 = 0.302$) is not large. Period effects within Plato are a secondary signal visible only once genre is controlled for, which is a direct methodological warning for studies that attribute compositional register differences to philosophical evolution.

6 Discussion

6.1 The Köntges hypothesis revisited

The evidence bears upon Köntges’s claim more narrowly than Table 1 alone would suggest. At the level of whole works entropy does fall with corpus size, and falls significantly; upon that evidence Köntges is right. What separates the two hypotheses is the within-work analysis, where corpus size is fixed by construction, and entropy still moves there — by roughly a fifth of the section mean across the *Phaedo*, and by a comparable margin within the *Metaphysics*. Whatever drives that variation, it is not the quantity of text available for disambiguation.

The two findings reconcile if Köntges’s claim is taken to cover two phenomena that behave differently. Frequency-based disambiguation of common forms does improve as corpora grow. The ambiguity carried by rare and philosophically loaded vocabulary appears instead to be conditioned by register, and there is no obvious reason to expect more data to resolve it.

6.2 What the three-method comparison tells us

The three DTM methods produce broadly consistent topic structures, with the quantified divergence captured by ϕ_{std} . The contribution of the probabilistic pipeline is not to overturn the baseline results but to produce, to

our knowledge, the first uncertainty-quantified topic model of the Platonic corpus, one in which the stability of each topic-word association is a measurable quantity rather than an assumption. Where previous computational work upon Greek style (e.g. Storey & Mimno, 2020) has treated corpus representations as fixed inputs, this framework treats the representation itself as a source of structured uncertainty. More broadly, it reframes preprocessing in digital humanities as a site where interpretive uncertainty can be measured and studied rather than a preliminary step to be standardised away. A classicist working with T12 (Form/Beauty) now has a quantitative basis for knowing which of its top words are stable across morphological realisations and which are sensitive to preprocessing choices, a distinction the standard approach does not make visible.

6.3 The Septuagint results in context

The Septuagint projection demonstrates both the reach and the limits of the approach. The Wisdom of Solomon result (T14 cosine = 0.8105) provides a quantitative measurement of Platonic vocabulary penetration into Hellenistic Jewish scripture, and the cosine-based disambiguation framework is essential to it: without that framework the T14 activation in Genesis (mean $\theta = 0.185$) would appear comparably strong, suggesting that the Genesis creation narrative reflects direct Platonic influence. The cosine analysis reveals that Genesis and Wisdom activate T14 through different vocabulary, and that only Wisdom’s activation reflects genuinely Platonic lexical overlap.

The negative result is also informative. Wisdom’s T10 activation shows an overlap in thanatological register that the pooled cosine places within the ambiguous band, and not the absorption of specifically Platonic soul-immortality discourse, as the absence of ἀποθνήσκω from Wisdom’s top T10 chunks confirms. The cosine framework is what makes this distinction measurable rather than merely arguable.

6.4 Limitations

Several limitations bear upon how the results should be read. The covariate regression is post-hoc and uses a simplified linear model: the R^2 values are descriptive summaries of association rather than causal estimates, and the genre and period covariates are correlated by construction — all late Platonic works are dialogues, all Aristotelian works are treatises — so the coefficients cannot fully separate their effects. A full structural topic model in the spirit of Roberts et al. (2014) would address this, at the cost of substantially greater computational complexity.

The Septuagint projection carries its own caveat. Projecting held-out texts into a trained topic space assumes that their vocabulary is sufficiently represented in the training lexicon; the 0.0% token drop rate across all four Septuagint works suggests adequate coverage, but the approach will underestimate philosophical overlap wherever genuinely distinctive Septuagint vocabulary falls outside the 1,336-lemma training vocabulary.

A second caveat concerns the uncertainty measure itself, and it is the one that most constrains what this paper can claim. As reported in Section 3.2, the published ϕ_{std} conflates morphological resampling with the fitter’s own initialisation variance, and the seed-only control recovers 98.8% of the measured spread. Had we not run that control, the headline figure would have been reported as morphological when it is largely not. A second error, independent of the first, ran in the opposite direction: coherence was originally computed on unaligned fits, which suppressed rather than inflated the association it was used to test. Both were errors of procedure invisible in the output, and neither would have been found without re-running the pipeline against a control it had not previously been asked to pass.

Finally, the Perseids Morpheus API is a rule-based parser with known gaps for rare forms and late Platonic vocabulary, and the posterior distributions it returns are engineering approximations rather than true Bayesian posteriors. Neural approaches offer a route past the rule-based ceiling — Singh, Rutten and Lefever (2021)

report gains on Ancient and Medieval Greek morphological analysis from BERT-style language modelling — though none that we are aware of yet exposes a calibrated token-level posterior of the kind this pipeline consumes. The most honest defence here is indirect: were Morpheus posteriors uninformative noise, ϕ_{morph} would be uniform across topics. The structured pattern we observe under the morphology-only control, with T12 nearly six times as unstable as T7 and the difference tracking semantic character, suggests that the uncertainty estimates are doing real work — and it is a stronger indication than the confounded measure supplied, since the same contrast stood at a ratio of 3.19 before estimation noise was removed.

7 Conclusion

This paper examined whether morphological ambiguity in Ancient Greek requires explicit probabilistic treatment or whether it diminishes with sufficient corpus size. The results suggest that ambiguity is semantically structured, conditioned by rhetorical register, and propagated into downstream topic modelling outputs in measurable ways.

These results form a logical chain rather than an independent list of contributions. The uncertainty-quantified pipeline — three DTMs encoding distinct epistemic stances toward morphological ambiguity — makes the ϕ_{std} measurement possible. That measurement, once separated from the fitter’s own initialisation variance, establishes that the topic structure is stable enough across morphological realisations to support the Septuagint projection with quantified confidence. And the projection demonstrates that the pipeline’s uncertainty estimates carry analytical value, since the cosine disambiguation framework is what distinguishes the Wisdom of Solomon’s genuine Platonic vocabulary overlap from Genesis’s superficially similar but lexically distinct T14 activation.

Across the corpus, uncertainty was neither uniform nor negligible. Topic instability varied systematically by semantic domain, within-work entropy varied substantially even where corpus size remained fixed, and the Septuagint projection showed that topic activation alone is insufficient to establish genuine lexical overlap. Together these findings indicate that deterministic preprocessing conceals meaningful uncertainty capable of influencing interpretation at multiple stages of the modelling pipeline.

For computational work upon Ancient Greek, and upon morphologically rich languages more generally, the practical conclusion is the narrower but more usable one: preprocessing uncertainty is large enough to measure, it is structured rather than random, and measuring it costs two additional loops over the corpus — one to resample the morphology, and one to establish how much of the resulting spread the fitter would have produced on its own.

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